

AI PRODUCT CASE STUDY

DECISION ADVANTAGE

IPCI OPERATIONS INTELLIGENCE PLATFORM

AI-POWERED HOSPITAL OPERATIONS INTELLIGENCE PLATFORM

DESIGNED AND DELIVERED END-TO-END



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DURING OPENAI BUILD WEEK**

Executive Summary

Decision Advantage is an AI-powered operations intelligence platform that helps hospitals detect pressure earlier and coordinate flow more effectively. Built end-to-end using synthetic data, it demonstrates responsible, human-in-the-loop AI in a high-stakes domain — spanning modelling, backend engineering, AI reasoning, UX design, cloud deployment, and governance.

My Role

I designed and built this platform solo, end-to-end, across every layer of the product:

- Owned the full system architecture, from data model to deployment
- Designed and implemented the synthetic operational modelling and uplift logic
- Built the backend (Python, FastAPI) and all operational/scenario logic
- Designed the AI reasoning workflows and integrated OpenAI for narrative generation and decision support
- Built the frontend (React/TypeScript) and designed the UX for clinical and operational roles
- Handled cloud deployment and continuous delivery (Render, GitHub)
- Designed the governance, explainability, and audit-trail approach, with GDPR / EU AI Act principles in mind

1. Overview

Decision Advantage is an AI-powered hospital operations intelligence platform designed to demonstrate how modern AI systems can support earlier, safer, and more coordinated decision-making in complex healthcare environments.

The platform integrates synthetic operational modelling, AI-driven reasoning, and enterprise-grade UX to deliver:

- Real-time operational summaries
- Flow Score v3 insights
- ED, Bed, DTOC, LOS, and Boarding pressure indicators
- Scenario forecasting (including Mixed Pressure)
- AI-generated priority actions
- Explainability and auditability
- Role-based operational views

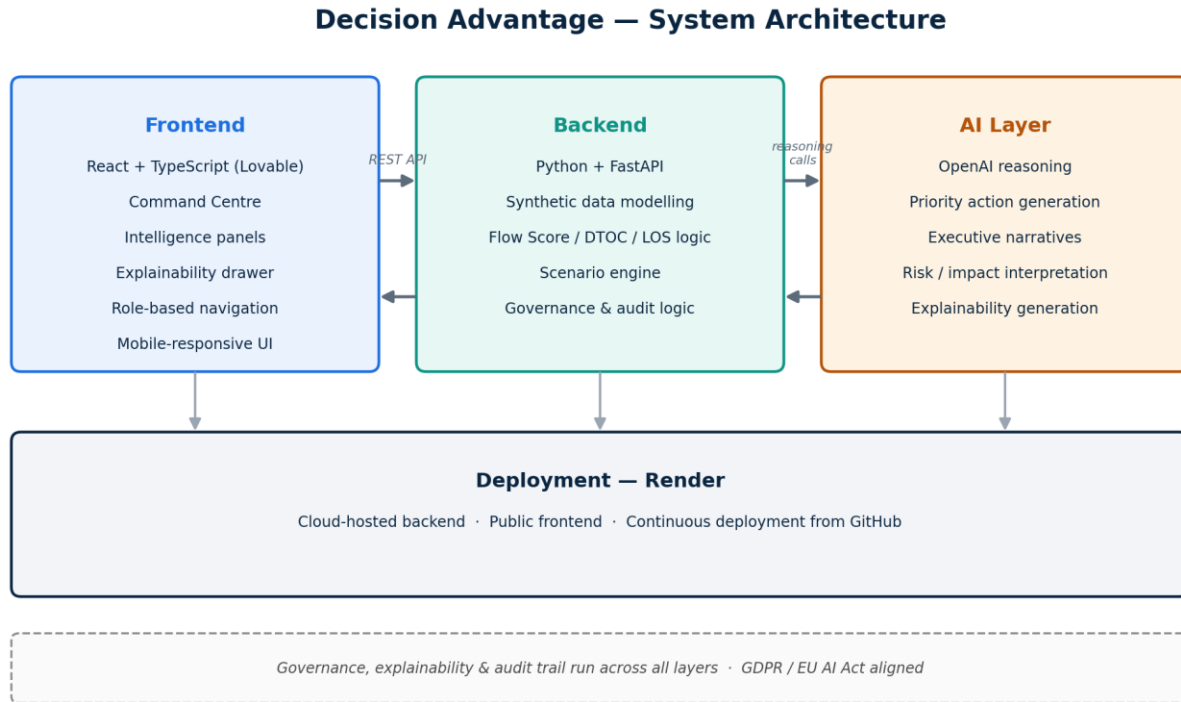
All data is fully synthetic and aligned with GDPR, DPA, and EU AI Act expectations.

2. Objectives

- Demonstrate how AI can enhance operational clarity in hospitals
- Model realistic pressure scenarios using synthetic uplift logic
- Provide actionable decision support with human oversight
- Deliver an enterprise-grade UX suitable for clinical and operational roles
- Showcase end-to-end AI product development capability

3. System Architecture

Decision Advantage is built as three cooperating layers — frontend, backend, and an AI reasoning layer — sitting on a single cloud deployment, with governance and explainability running as a cross-cutting concern rather than a bolt-on feature.



High-level architecture: request flow between frontend, backend, and AI reasoning layer, deployed on Render with continuous deployment from GitHub.

Frontend (Lovable / React + TypeScript)

- Command Centre
- Advanced Intelligence dashboards
- Priority Action engine
- Explainability drawer
- Audit Trail
- Leadership Summary
- Role-based navigation
- Mobile-responsive enterprise UI

Backend (Python + FastAPI)

- Synthetic data modelling
- Operational uplift simulation
- Flow Score v3 logic
- DTOC, LOS, Boarding modelling

- Scenario engine (including Mixed Pressure)
- AI reasoning workflows
- Governance & audit logic

AI Layer (OpenAI)

- Narrative generation
- Priority action reasoning
- Executive summaries
- Risk/impact interpretation
- Explainability generation

Deployment (Render)

- Cloud-hosted backend
- Public frontend
- Continuous deployment from GitHub

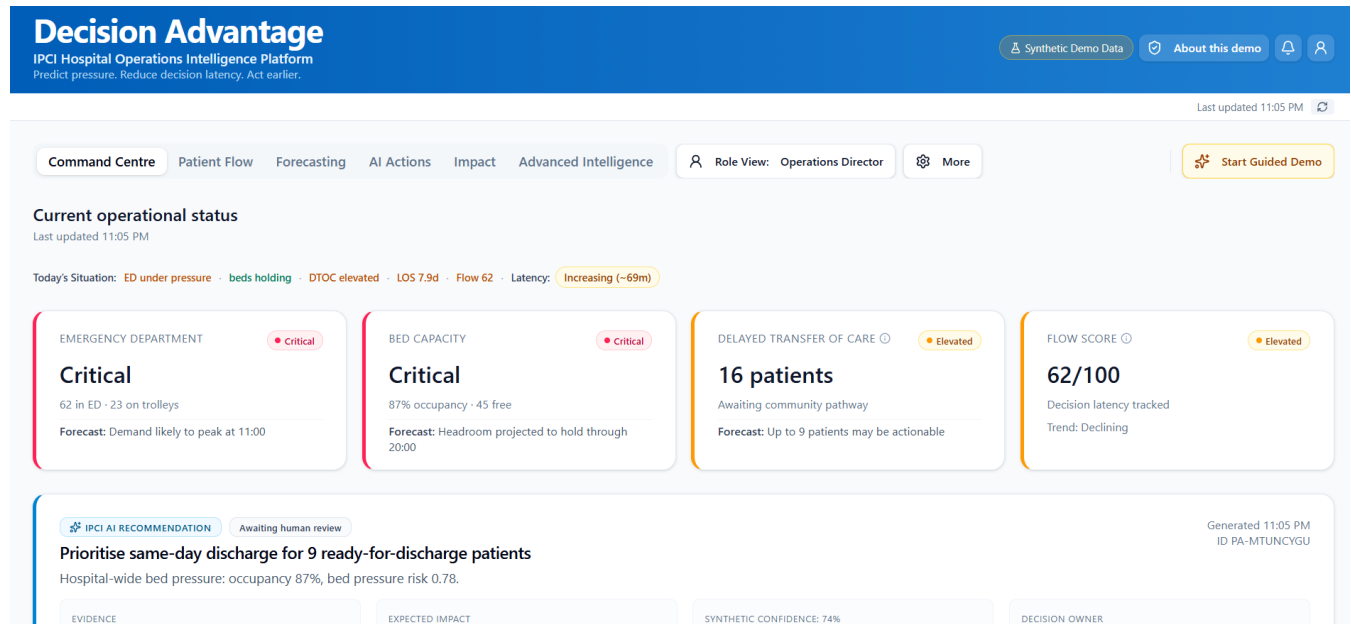
How data flows through the system

1. **Frontend → Backend:** The React/TypeScript frontend renders role-specific views and sends requests to the backend over a REST API.
2. **Backend logic:** FastAPI computes operational metrics (Flow Score, DTOC, LOS, boarding) from the synthetic operational model, and calls the AI layer when a scenario requires reasoning, narrative generation, or a priority action.
3. **Backend → AI layer:** OpenAI is used to interpret the current operational signals and generate a recommendation, an evidence summary, a confidence score, and — where relevant — an executive narrative.
4. **AI layer → Backend → Frontend:** AI-generated recommendations return to the backend with structured evidence and confidence, are logged to the audit trail, and are surfaced to the frontend for human review, approval, or dismissal.
5. **Governance layer (cross-cutting):** Every step above is logged for explainability and auditability — signals, assumptions, and forecast horizon are stored alongside each decision, aligned with GDPR, DPA, and EU AI Act expectations.

4. Key Features

Command Centre

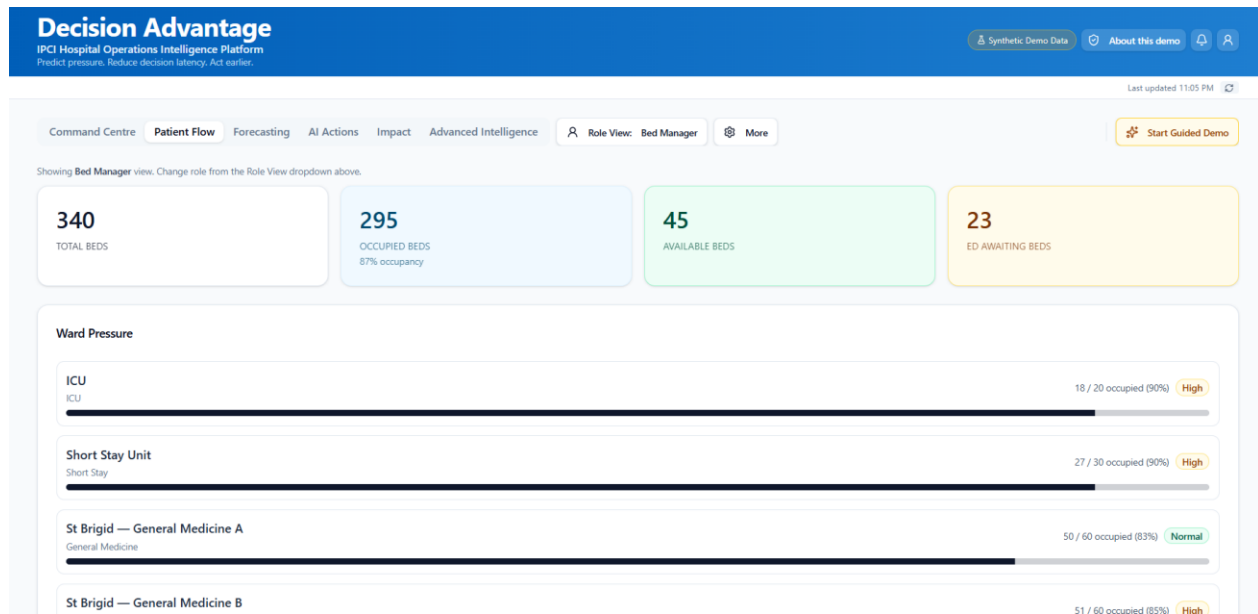
- Compact operational status
- Decision latency modelling
- Colour-coded severity indicators
- Mobile-friendly layout



Command Centre — Operations Director view. Real-time operational status across ED, bed capacity, DTCO and Flow Score, with an AI-generated priority recommendation awaiting human review.

Advanced Intelligence Panels

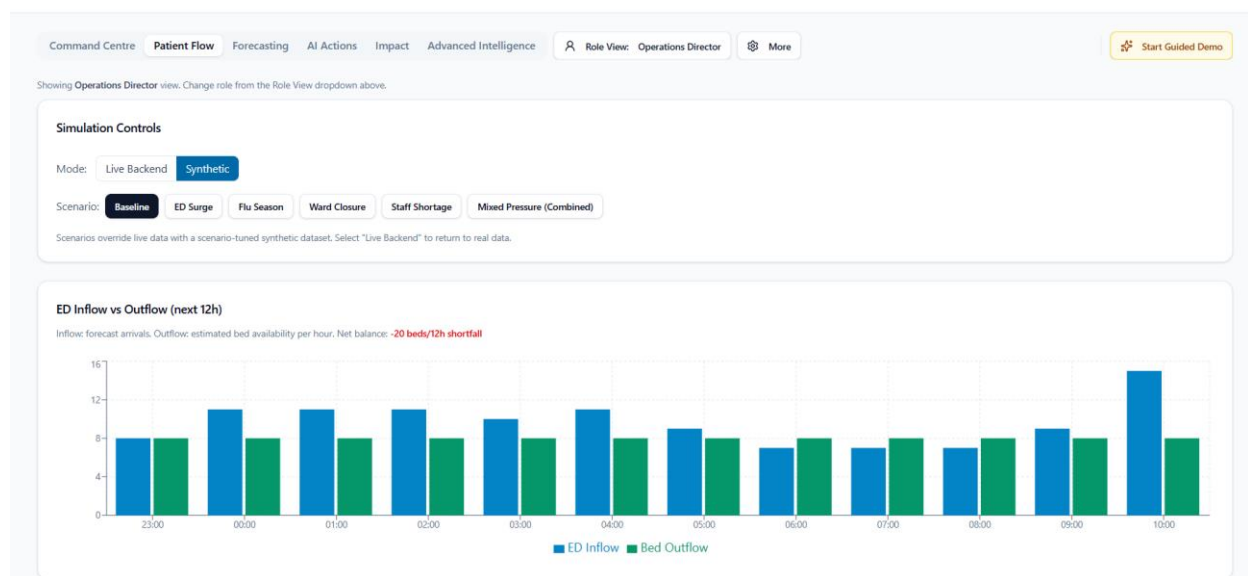
- Flow Score v3
- ED Flow
- Inpatient Flow
- Discharge Flow
- DTCO
- LOS > 14 days
- ED Boarding
- Human Impact



Patient Flow — Bed Manager view. Ward-level bed pressure broken down by unit (ICU, Short Stay, General Medicine).

Mixed Pressure Scenario Engine

- Combined ED surge
- DTOC uplift
- LOS increase
- Staffing strain
- Boarding pressure
- Realistic synthetic deltas



Forecasting — Scenario Engine. Simulation controls for scenario testing (ED Surge, Flu Season, Ward Closure, Staff Shortage, Mixed Pressure), with a 12-hour ED inflow vs. bed outflow forecast.

AI Priority Action Engine

- Action recommendations
- Evidence and confidence
- Operational signals
- Human-review workflow

Explainability Drawer

- Signals
- Assumptions
- Uncertainty
- Risks
- Forecast horizon

Audit Trail

- Pending → Completed → Assigned → Dismissed (Synthetic)
- Full decision lifecycle
- Governance-aligned

Leadership Summary & Briefing

- Executive-ready narrative generation
- Scenario interpretation
- Operational impact summary

Role-Based Navigation

- ED Nurse
- Bed Manager
- Inpatient Flow
- Hospital Manager
- Operations Director (default)

5. Technologies Used

- Python — modelling & logic
- FastAPI — backend API
- Pydantic — typed data contracts for reliability and validation
- OpenAI (GPT + Codex) — reasoning, code generation & narrative generation
- Lovable (React + TypeScript) — frontend
- Render — cloud deployment
- GitHub — version control
- Modern UX patterns — enterprise healthcare design

6. Skills Demonstrated

- AI systems design
- Synthetic data modelling
- Operational uplift simulation
- Full-stack development (Python, FastAPI, React/TS)
- AI explainability & governance
- Scenario modelling & forecasting
- Cloud deployment
- Enterprise UX/UI design
- Human-in-the-loop decision systems

7. Challenges & Trade-offs

Building an end-to-end AI product solo, across modelling, backend, AI reasoning, and enterprise UX, surfaced a number of design decisions worth calling out explicitly.

1. Synthetic data realism vs. simplicity

Hospital operational data (bed occupancy, DTOC, LOS, boarding) needed to feel realistic enough to support meaningful scenario testing, without access to real patient data. The trade-off was between a fully generative statistical model (more realistic, harder to explain and validate) and a rules-based uplift model tuned to published Irish HSE-aligned KPI ranges (more transparent, easier to justify to a non-technical audience, but a simplification of real-world variability). I chose the rules-based approach so that every number in the platform could be traced back to an explainable assumption — a deliberate trade-off of statistical sophistication for auditability.

2. AI confidence vs. actionability

Early versions of the Priority Action Engine surfaced AI recommendations without confidence scores, which made them look more authoritative than they should. Adding explicit confidence percentages, evidence panels, and a mandatory human-review step reduced the apparent authority of the AI output — a conscious trade-off of "how impressive it looks" for "how appropriate it is for a clinical decision-support context."

3. Depth of explainability vs. cognitive load

The Explainability Drawer (signals, assumptions, uncertainty, risks, forecast horizon) could easily have become a dense wall of technical detail. The challenge was surfacing enough reasoning for a decision-maker to trust and audit a recommendation, without burying the actionable summary underneath it. This was solved by separating the Command Centre (compact, decision-first) from the Explainability Drawer (opt-in, detail-first) as distinct UI layers.

4. Role-based scope vs. engineering overhead

Supporting five distinct role views (ED Nurse, Bed Manager, Inpatient Flow, Hospital Manager, Operations Director) meant the same underlying data needed different framing, granularity, and default views per role. Rather than building five separate views, a single configurable navigation and panel-visibility layer was built on top of one data model — more upfront engineering effort, but avoiding long-term duplication as features were added.

5. Scope boundary: demonstration vs. production system

The platform is intentionally built on synthetic data and is explicitly labelled as such throughout the UI ("Synthetic Demo Data", "About this demo"). The trade-off was between building toward a narrower, production-ready integration (e.g. a single hospital's real EHR feed) versus a broader, fully-featured demonstration of the AI product thinking across the full operational surface. Given the goal of showcasing end-to-end AI product capability rather than shipping a deployable clinical system, the latter was the right scope — with governance and explainability designed to the same standard a production system would require.

Together, these decisions reflect a consistent design principle across the project: prioritise transparency, human oversight, and auditability over raw model sophistication or visual polish — appropriate for AI operating in a high-stakes, regulated domain.

8. Outcome

Decision Advantage demonstrates how hospitals can leverage synthetic modelling and AI reasoning to detect pressure earlier, coordinate flow more effectively, and support human decision-makers with clearer, more actionable insights.

The project reflects capability in end-to-end AI product development, from architecture and modelling to UX, deployment, and governance.

Links

- **Live Platform:** operah-care.lovable.app
- **Project submission:** <https://devpost.com/software/decision-advantage-ipci-operations-intelligence-co-pilot>